Lecture 2
CO_2 Transport and Storage

Yongchun Zhao

State Key Laboratory of Coal Combustion
Huazhong University of Science and Technology

Outline

- CO_2 transport
- CO_2 storage
- International CO_2 storage activities
- CO_2 storage R&D in China

Storage overview-Site options

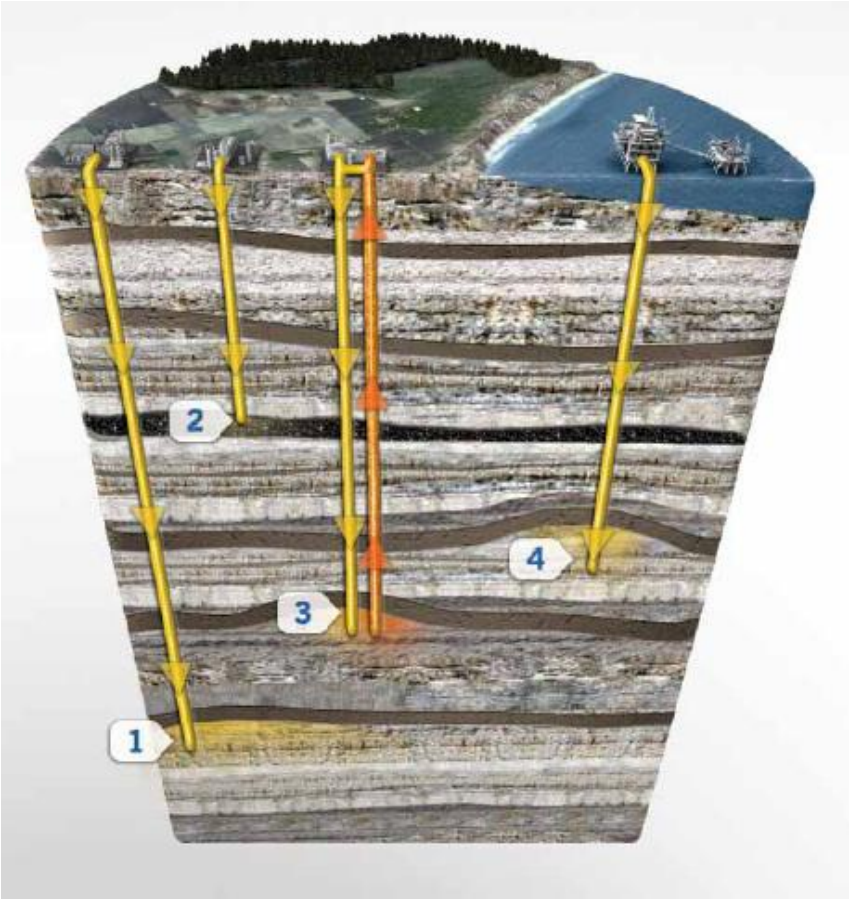
Globally, approximately 25 Mtpa of anthropogenic (human made) CO_2 can be stored in geological reservoirs by currently operating LSIPs. Of this, incidental storage associated with EOR activity accounts for more than 90 percent.

① Saline formations盐水层

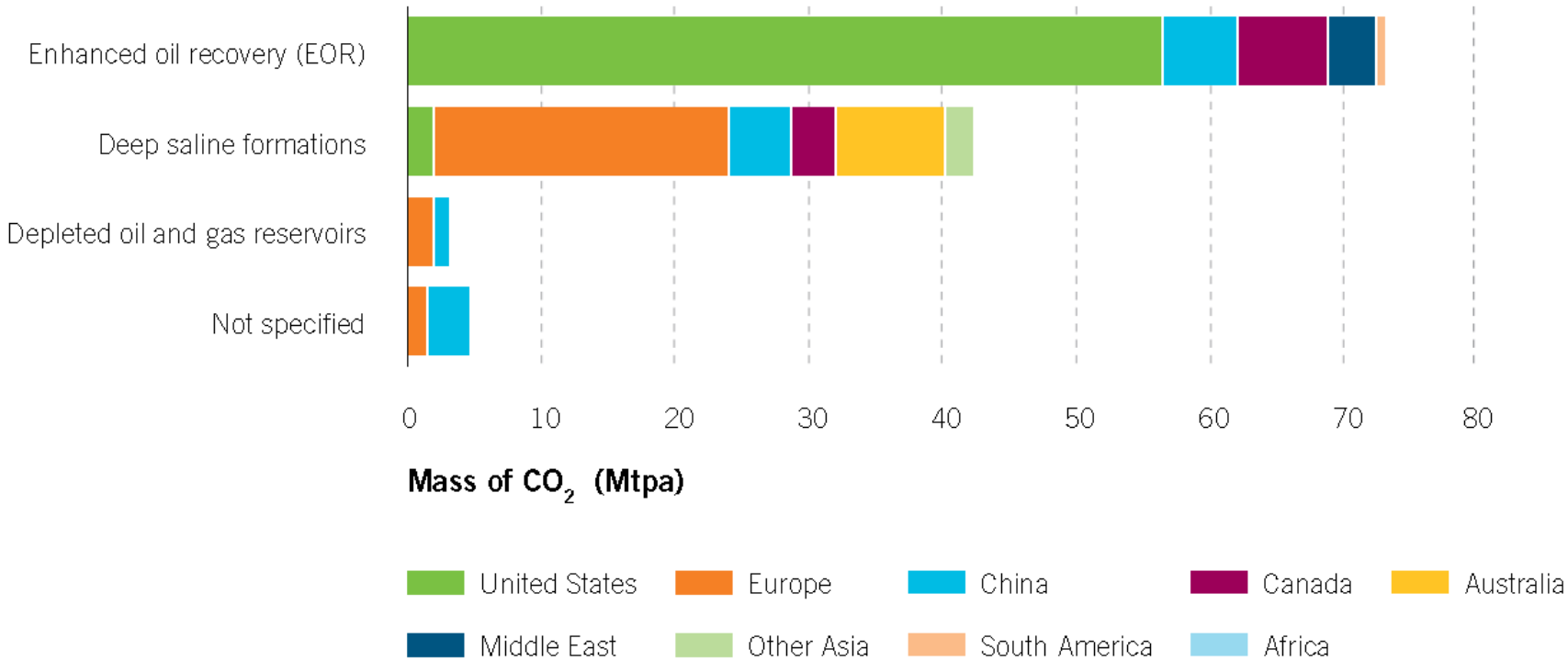
② Injection into deep unmineable coal seams 不可采煤层 or ECBM

③ Use of CO_2 in enhanced oil recovery驱油

④ Depleted oil and gas reservoirs枯竭油气层

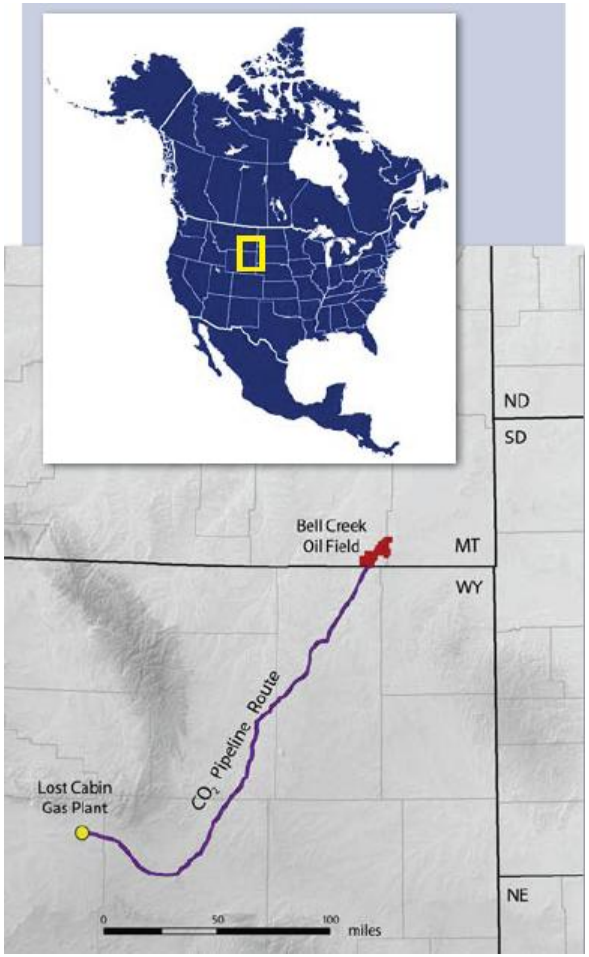


LSIP storage types according to region and country



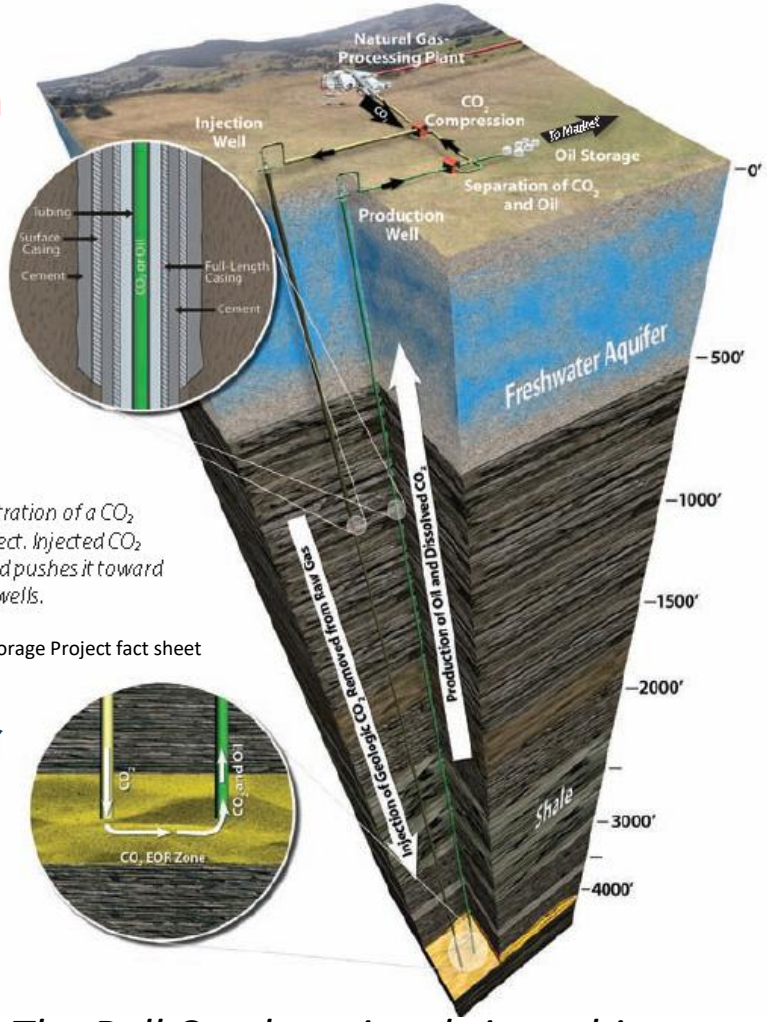
The distribution of CO_2–EOR operations has been concentrated in North America, in particular Texas, Oklahoma, and New Mexico, but opportunities are being recognised more broadly now in Brazil, Mexico, the Middle East, North Sea, and China

Bell Creek Integrated CO_2 EOR and Storage



Bell Creek Project Benefits 35 million barrels of incremental oil ……

. . . millions of tons of CO2 safely in storage.



Simplified illustration of a CO_2 flood EOR project. Injected CO_2 thins the oil and pushes it toward oil-producing wells.

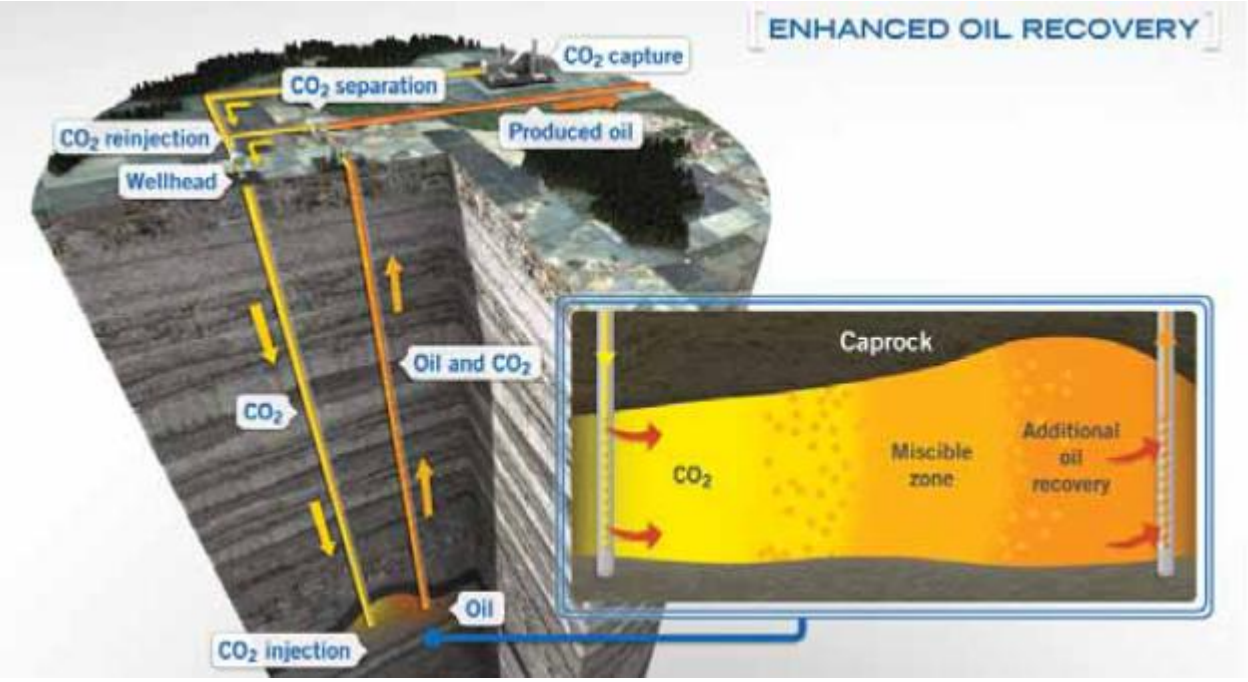
Source: Bell Creek Integrated CO2 EOR and Storage Project fact sheet

全流程：捕获、输运、埋存、EOR、监测、公众

CO_2 will be transported 232 miles by pipeline from the Lost Cabin gas-processing facility in central Wyoming to the Bell Creek oil field. At this point, CO_2 will be injected into the oil-bearing rock of the Muddy Formation sandstone.

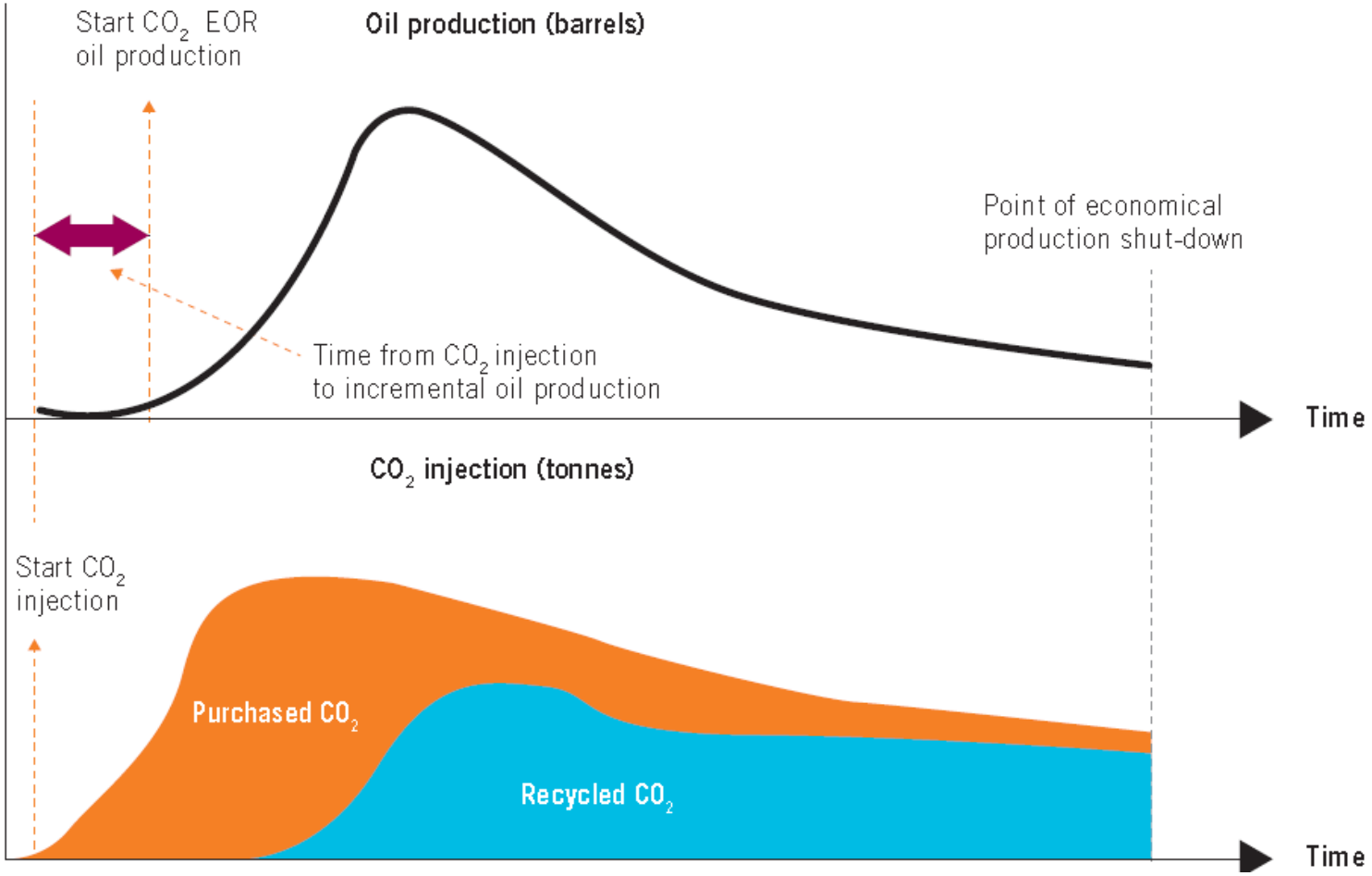
The Bell Creek project brings this approach of integrated EOR storage and MVA to a pioneering project in the northern Great Plains.

The CO_2 enhanced oil recovery process



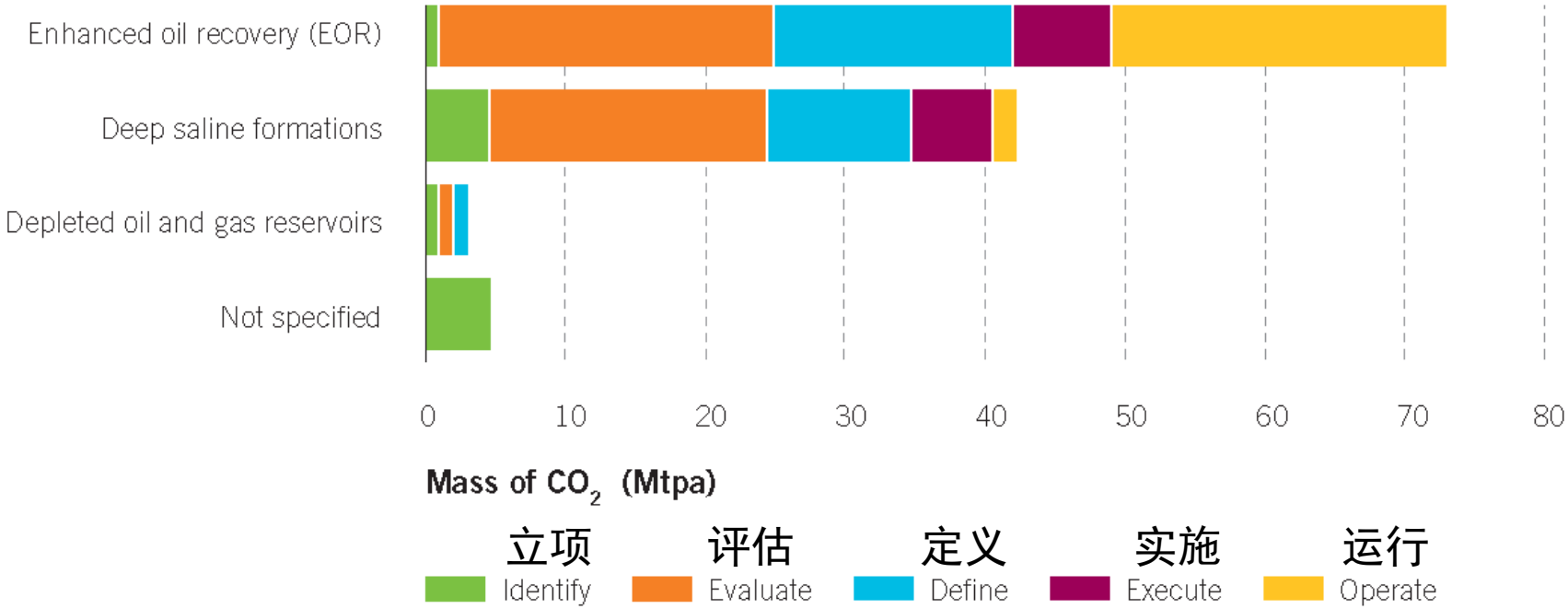
The CO_2 is injected as a compressed liquid that dissolves into the oil. Conversely the oil dissolves into the dense CO_2. Depending on the pressure and temperature of the reservoir and oil type, the CO_2 and oil may mix to become a single phase (miscible). Miscibility causes the oil to swell slightly and become less viscous, so that it flows within and through the reservoir pores more easily. The oil then comes to the surface, the well pressure decreases, and the once miscible CO_2 un-mixes from the oil.

Relative increase in recycled CO_2 being used compared to newly purchased CO_2 during the course of a CO_2–EOR operation



Storage potential associated with CO_2–EOR

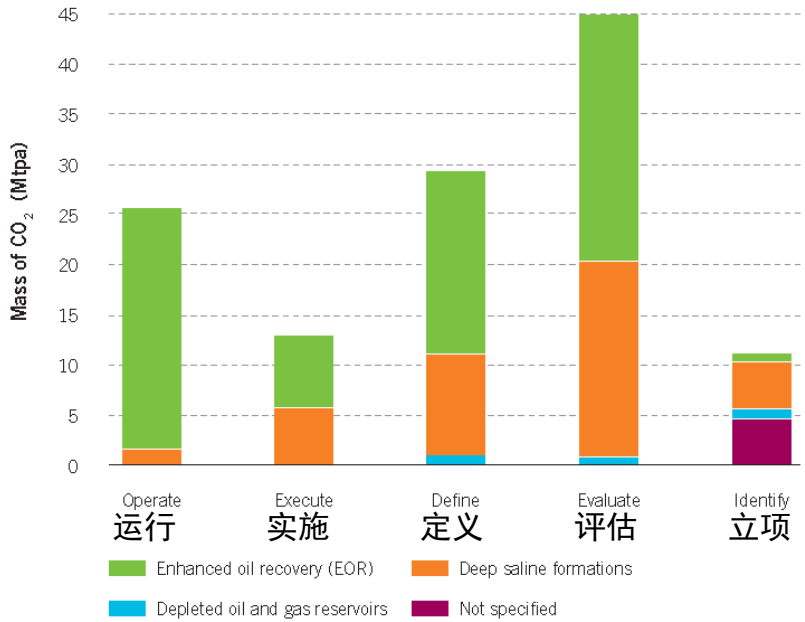
Potential mass of CO_2 stored by 65 LSIPs according to storage type and project lifecycle stage: approximately 122 Mtpa.

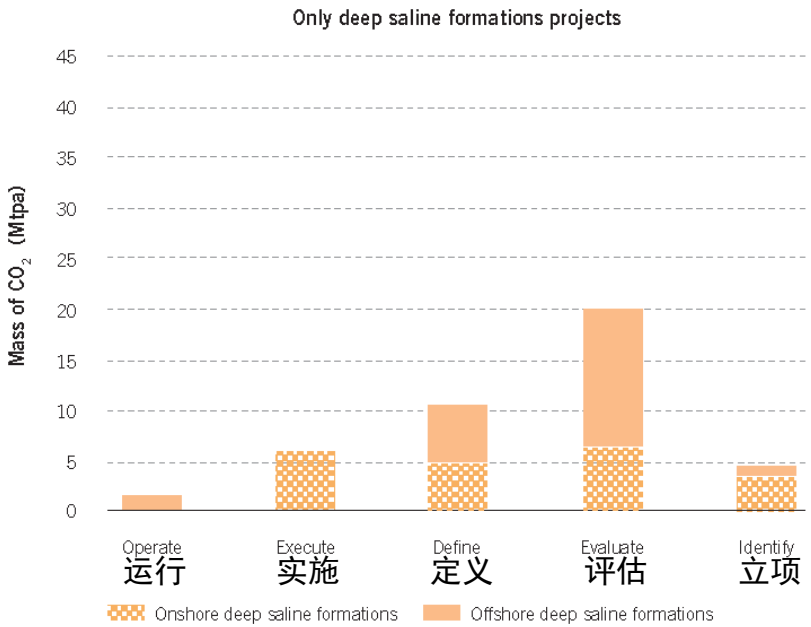


More than 20 billion tonnes of CO_2 storage potential is estimated to be associated with CO_2–EOR in North America. Although these numbers are preliminary, they indicate the significant storage potential associated with CO_2–EOR, which could be important for future CCS development.

Dedicated storage activities(Non-EOR)

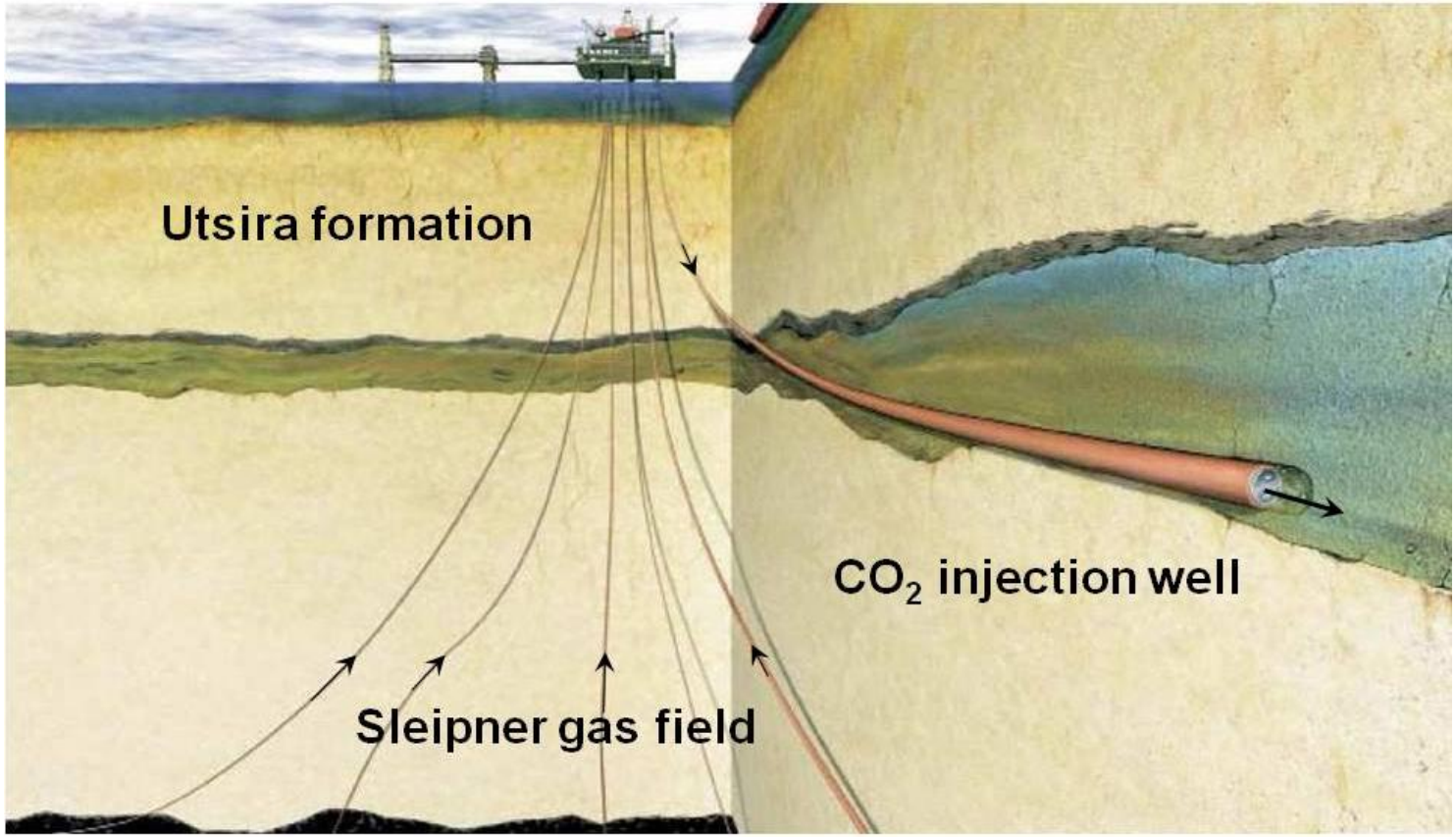
Mass of CO_2 that could be stored by LSIPs





Since 2012, operational large–scale storage projects targeting saline formations (i.e. non-EOR activities) injected and stored about 1.5 Mt of CO_2. Similar projects in construction (Execute) should increase CO_2 injection into saline formations by about 5.8 Mtpa to more than 7 Mtpa in 2015, thus quadrupling the amount currently stored in saline formations.

Sleipner Project in Norway-**Operation**



The source of CO_2 is the Sleipner West gas field where it is separated from natural gas and then injected into a saline formation (called Utsira formation) 800 m below the seabed of the North Sea. Sleipner is the world's first commercial–scale dedicated storage project. CO_2 is currently injected at a rate of about 0.9 Mtpa and, to date, more than 14 Mt of CO_2 has been injected and stored.

In Salah CO_2 Storage project-**Operation**

Since 2004, CO_2 associated with processing natural gas has been injected into 1.9 km deep sandstones at the Krechba gas field. More than 3.8 Mt of CO_2 had been injected at an average rate of 0.5 Mtpa.

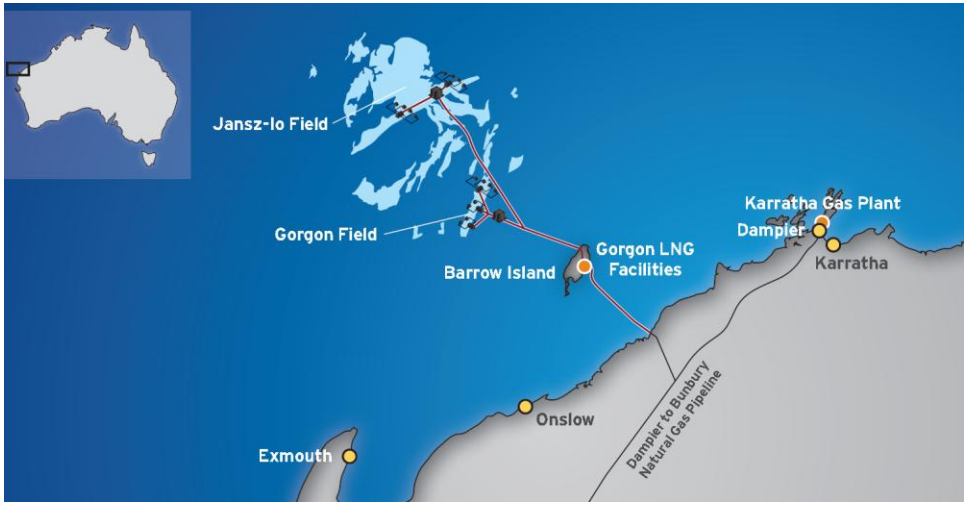
Storage formation overview:

- Injection formation: 20 metres-thick Carboniferous sandstone(石炭纪砂岩)
- Porosity and permeability: 13-20% porosity and 10mD permeability (very similar to potential deep saline CO_2 storage formations in many parts of the world)
- Injection depth: 1950m below ground with 13-20% porosity
- Wells: three, state-of-the-art, horizontal wells were drilled to inject the CO_2

Gorgon Injection Project in Western Australia

Gorgon Project Plant Site (February 2013).

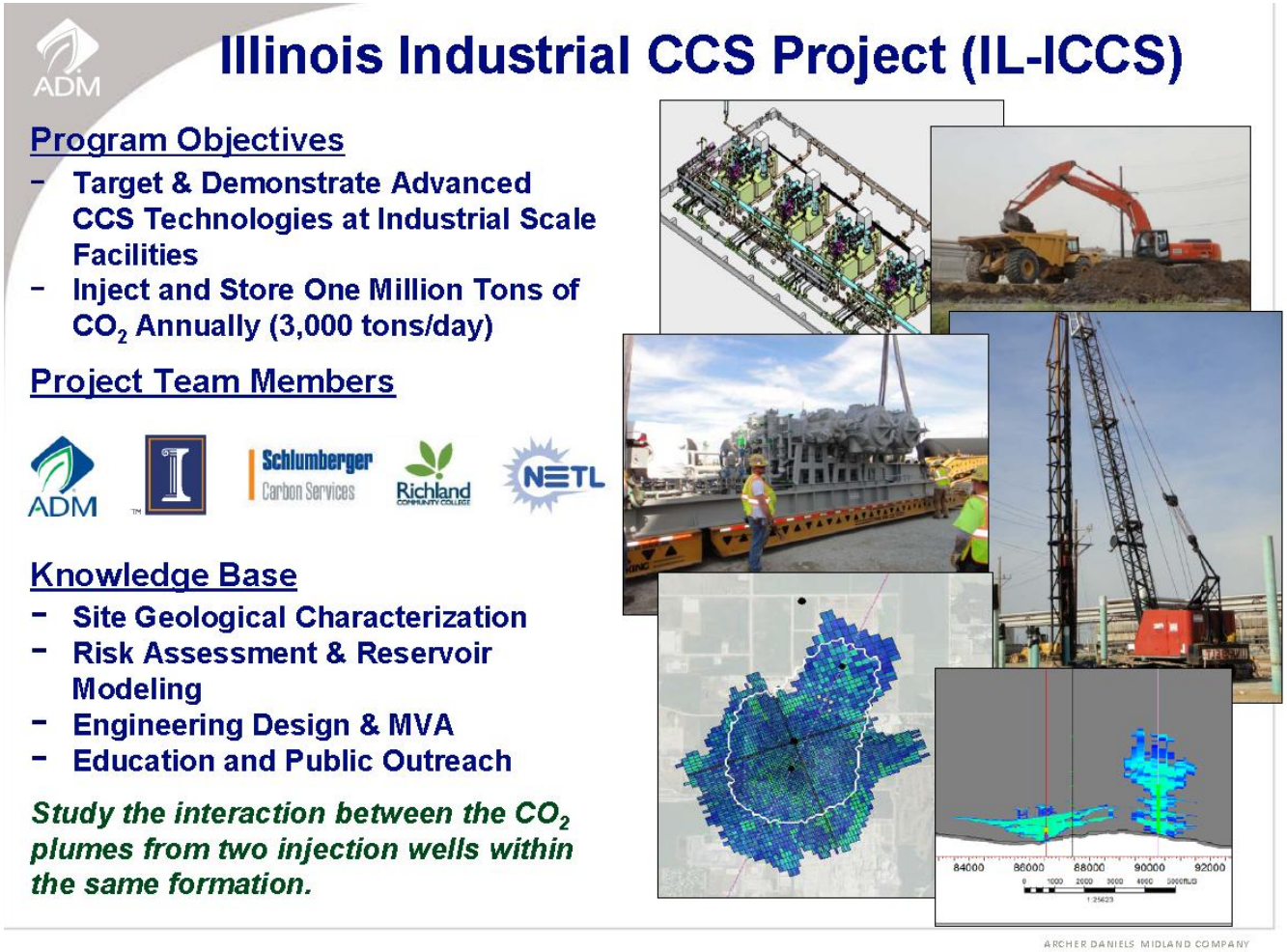
- Operational in 2015, Gorgon will be the world's largest CO_2 injection project.
- The project will inject between 3.4 and 4.1 Mtpa of CO_2 into sandstones at the Dupuy Formation about 2.3 km below the surface of Barrow Island.
- Barrow Island is located about 55 km off the coast of Western Australia, although the injection project is considered onshore storage.



Quest project in north-central Alberta, Canada

Quest is a fully-integrated CCS project that will capture, transport, inject and store CO_2. It was proposed by Shell on behalf of the Athabasca Oil Sands project – a joint venture between Shell Canada, Chevron Canada and Marathon Oil. Shell announced in September 2012 that a final investment decision had been made and that the project, having received essential federal and provincial regulatory approvals, had begun construction. The total cost of the project is estimated at $1.35 billion. The Quest venture is aiming for commissioning and injection to begin in 2015. Quest will store up to 1.1 Mtpa of CO_2 in the Basal Cambrian Sands about 2.2 km deep in the Alberta Basin and intends to inject 27 Mt over the life of the project.

The Illinois Industrial CCS Project



From mid-2014, Illinois ICCS intends to inject 1 Mtpa of CO_2 into the Mt Simon sandstones in the Illinois Basin at a depth of nearly 2.2 km. Illinois ICCS is notable in that it includes a significant public outreach and education program, including an association with the nearby National Sequestration Education Center.

FutureGen 2.0 Project（cancelled）

- **The total capital cost for the project is approximately $1.65 billion.**
- **Construction will begin in 2014 and the project will be online in 2017.**

美国伊利诺伊州的梅勒多西亚能源中心电厂

- *Carbon dioxide (CO2) capture: Retrofitting an existing coal-fueled power plant in Meredosia, Illinois with cutting edge oxy-combustion clean coal technology*
- *Transport: Construction of a CO2 pipeline from Meredosia to a CO2 storage facility*
- *Storage: Development of a CO2 storage facility in northeastern Morgan County, Illinois*
- *Facilities: Construction of a visitor, research and training center*

http://futuregenalliance.org/futuregen-2-0-project/

Storage activities in developing countries

Regional storage assessment in Southeast Asia

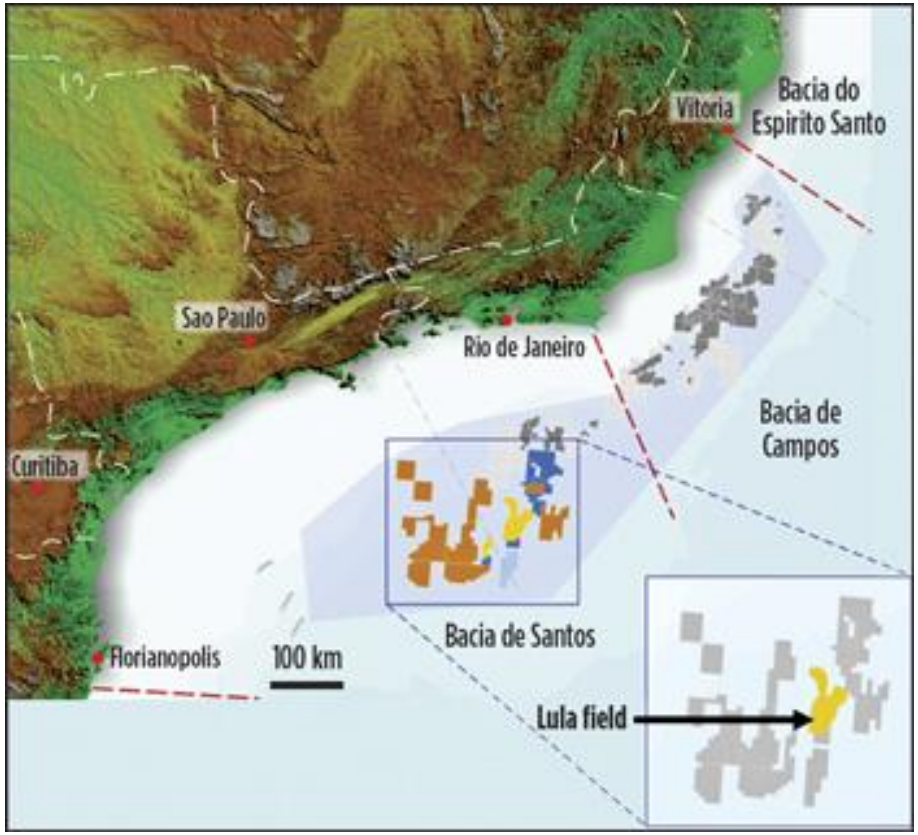
- The project is designed to support existing CO_2 geological storage activities and kick-start the implementation of storage mapping in member countries that do not have such activities.
- The aim is to provide a forum for knowledge sharing among the member countries and develop a guideline for national CO_2 storage mapping, along with a CCOP CO_2 storage atlas.
- Injectivity: can we put the CO_2 into the rock?
- Containment: can we keep the CO_2 in the rock?
- Storage capacity: what volume of CO_2 can the rock hold?

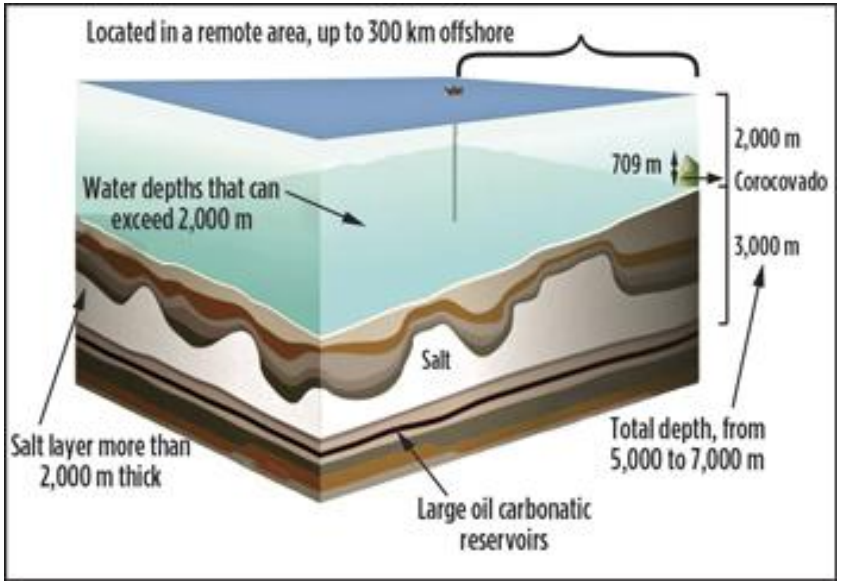
Map of CCOP countries

Coordinating Committee for Geoscience Programmes in East and Southeast Asia (CCOP)

Storage activity in Brazil

Brazil initiated the world's first offshore CO_2–EOR project in 2013, which also joined the ranks of operational LSIPs. In June, Petrobras and its partners commenced Phase 1A of the project, which is to inject and monitor 0.7 Mtpa of CO_2 into the Lula oil field.





The field is located in the pre-salt Santo Basin, which lies 300 km offshore in water 2,150 m deep, making it the deepest CO_2 injection well in operation.

Storage activity in China

Carbon dioxide utilisation continues to be an important part of the CCS discussion in China, and is often considered necessary to partially offset its cost. Three of four state–owned petroleum companies have CO_2–EOR pilots in operation:

- PetroChina's Jilin Oil Field EOR Project (Phase 1) commenced in 2008 and is now storing about 160,000 tpa of CO_2
- Sinopec's Shengli Oil Field EOR Project (Phase 1) captures and stores around 40,000 tpa of CO_2
- Yanchang Jingbian CCS Project (Phase 1) captures around 50,000 tpa of CO_2 for injection into sandstones at the Yanchang Oil Field in the Ordos Basin
- in Qinshi County, Shanxi Province, CO_2 has been injected since April 2010 as part of an enhanced coal-bed methane project.

Storage in deep saline formations and depleted oil and gas formations continues to gain relevance in China, with four LSIPs reporting this as the potential storage option for their projects, up from two the previous year.

- The Shenhua Ordos CTL Project (Phase 1), which aims to capture and store 100,000 tpa of CO_2, is among the most advanced projects in this category. It began injecting CO_2 in a saline reservoir within the Ordos Basin in 2011.

Storage activity in China

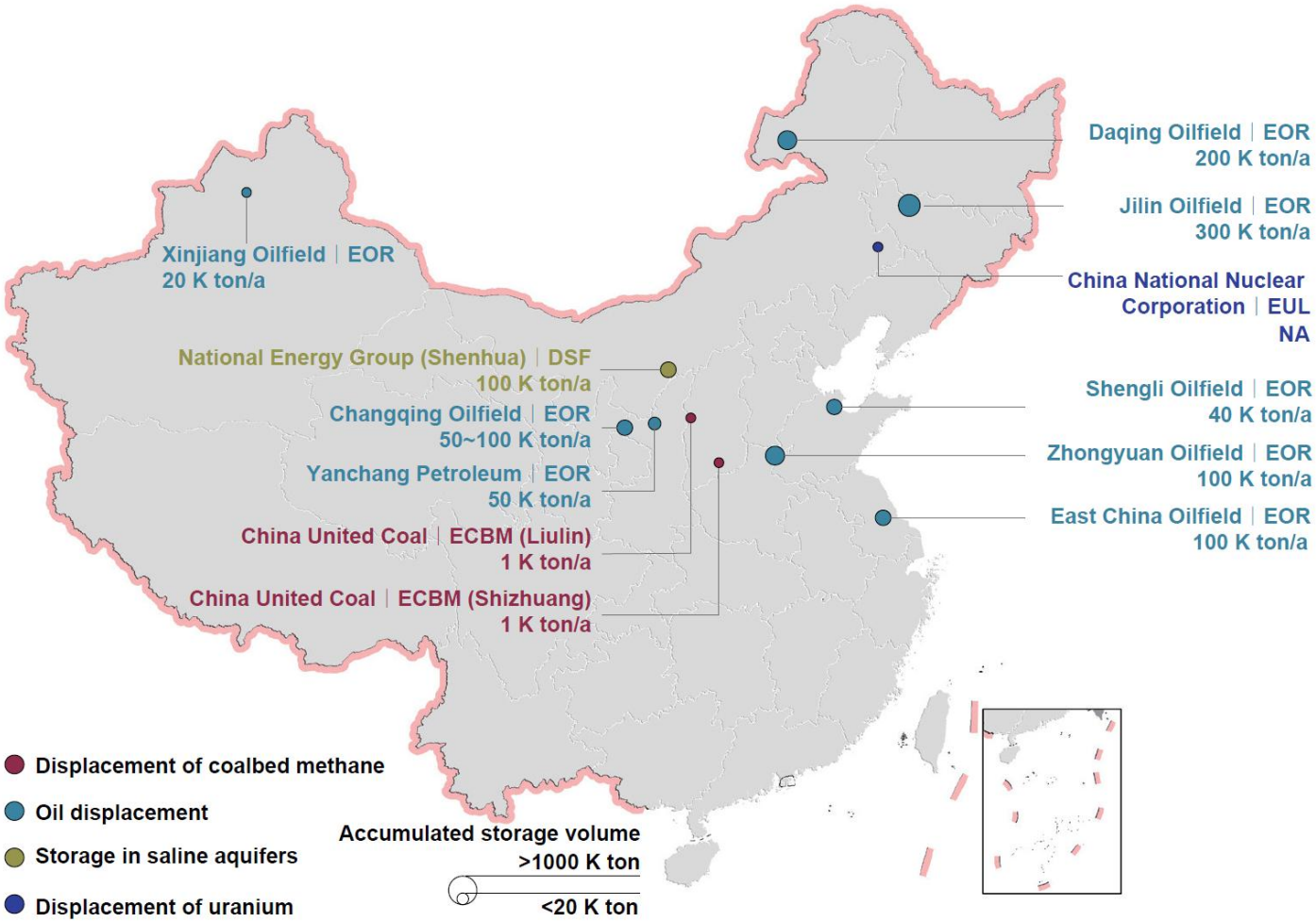
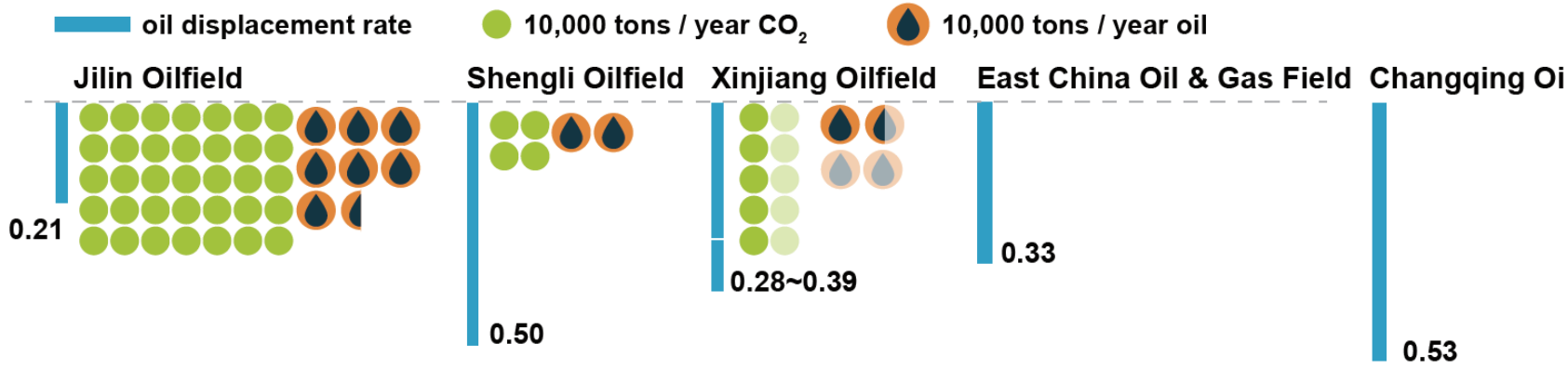


Fig. 10 Distribution of geological utilization and storage projects in China

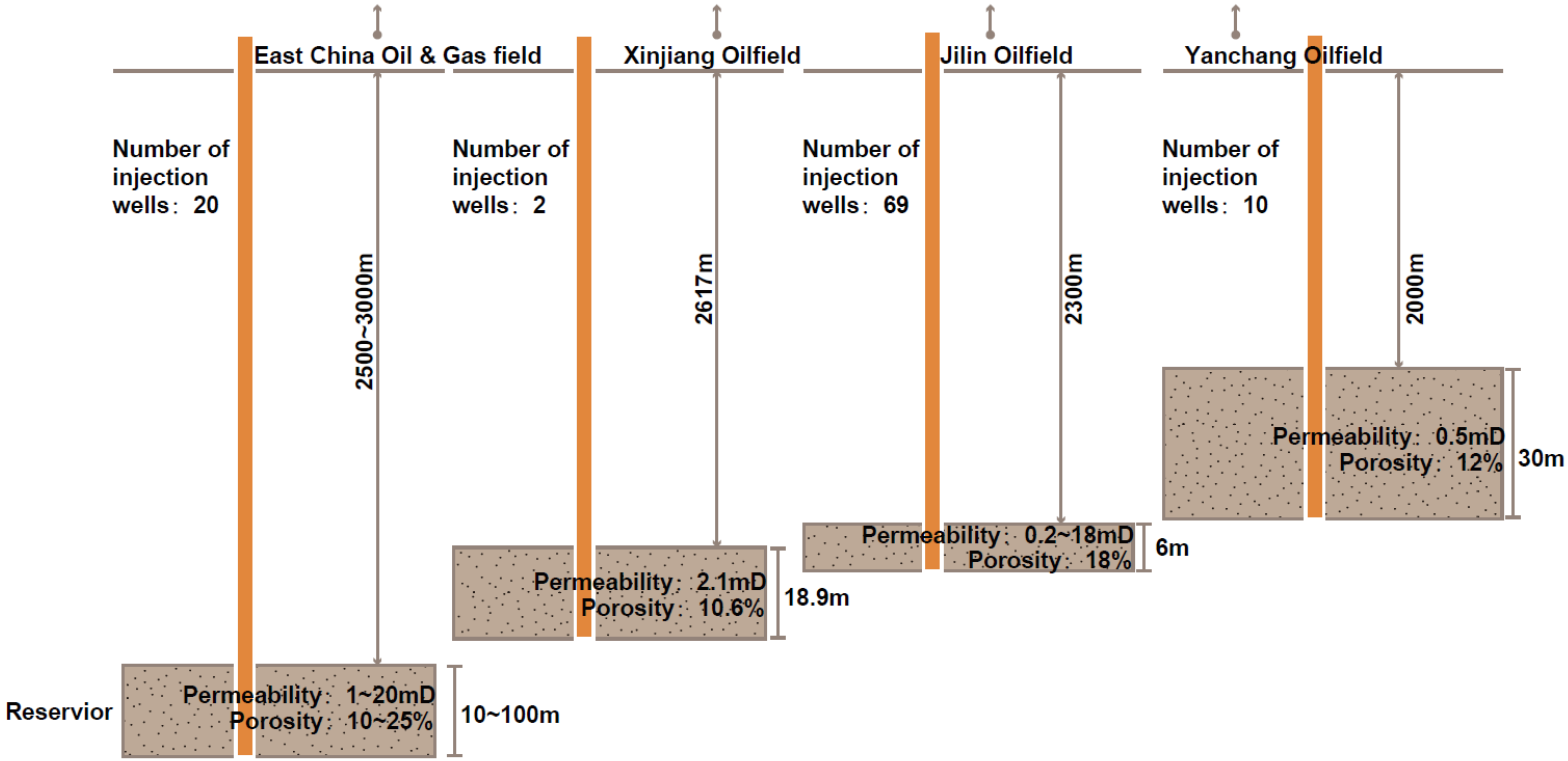
Carbon dioxide enhanced oil recovery (CO2-EOR) in China



CO_2 injection well and producing well in Jilin Oilfield, CNPC

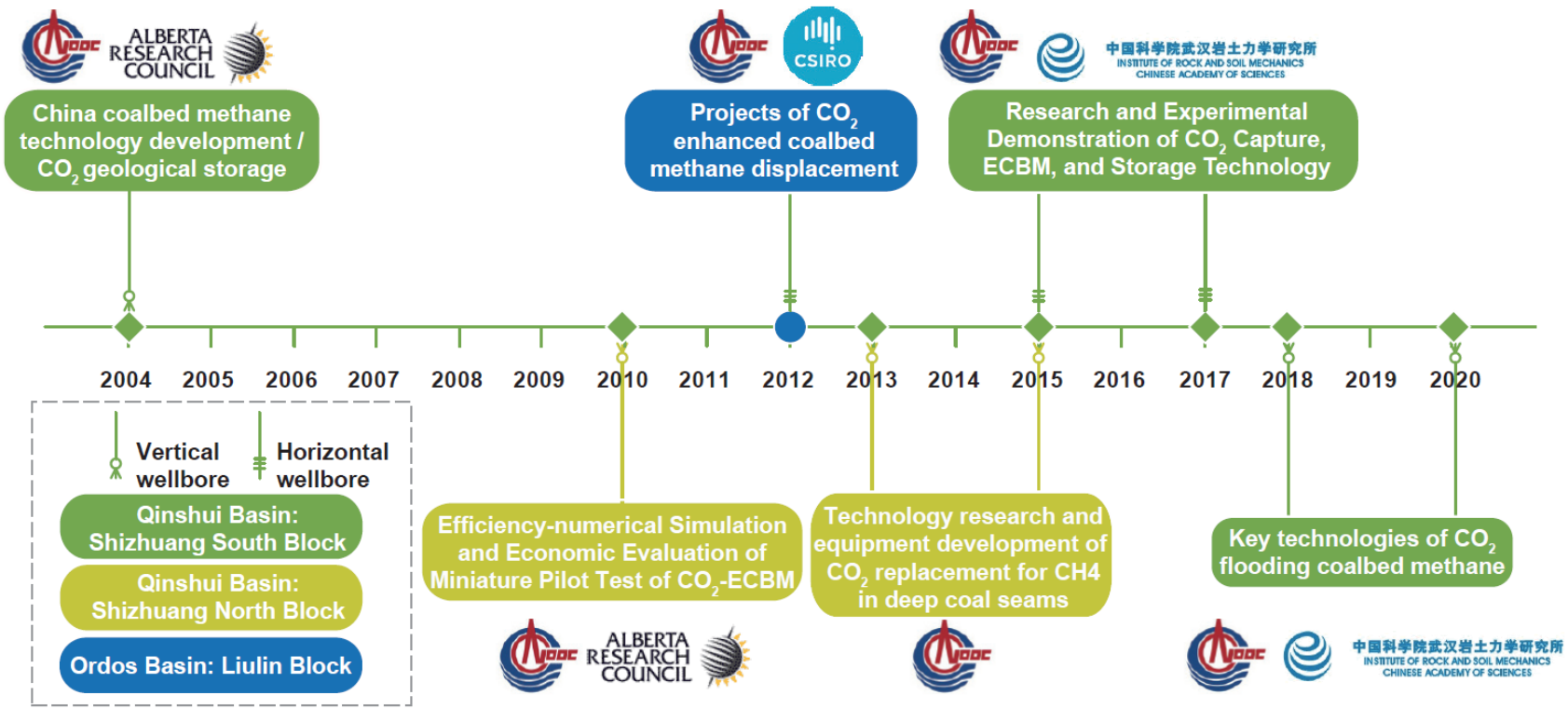
Carbon dioxide enhanced oil recovery (CO2-EOR) in China

Wellhead injection phase	Liquid phase	Liquid phase	Super critical phase	Liquid phase
Maximum Wellhead Injection Pressure	30MPa	19.7MPa	20MPa	13MPa
Average wellhead injection pressure	15MPa	16.1MPa	18MPa	9MPa
Wellhead injection temperature	1~20℃	-17℃	38℃	-20℃
CO_2 concentration	98%	99.99%	99.9%	99.8%



Basic parameters of a typical CO2-EOR project

CO2 - Enhanced coal bed methane recovery (CO2 -ECBM) in China



Pilot/demonstration trials of ECBM in China

Saline aquifer storage in China

Transport:
Road, tranker , 13km

Purified, compressed
CO_2 purity: 99.99%

Capture: Shenhua Coal-to-Liquid Chemical Co., Ltd. Ordos Branch
CO_2 purity: 88.8%

Buffer area,
Pressure, high temperature, pump injection

Injection well
Multi-layer injection, layered monitoring

This project started injection on May 9, 2011, and completed in 2015. By the end of 2015, the injection target of 300,000 tons had been achieved.

01.海上 CCUS 项目示范

我国东部沿海地区碳排放高、陆上碳封存资源少，而将 CO_2 运输至西部地区封存成本过高。依托海上 CCUS 项目的发展，可为沿海高排放企业提供降碳新思路。相关示范项目正在开展：

- 海上 CO_2 回注项目。2023 年 6 月，中国海油在恩平 15-1 油田投用了我国海上首个碳捕集与封存示范工程，累计安全运行近 13000 小时，回注二氧化碳超 18 万吨。
- 海上 CO_2 驱油封存项目。2025 年 3 月，我国珠江口盆地的恩平 15-1 平台开钻，采用“老井新用+分层注气”模式，预计 10 年累计规模化向海底地层注入二氧化碳超 100 万吨，同时提高油藏采收率驱动原油增产。

海上 CCUS 示范工程

02.陆上咸水层封存项目

- 陕煤集团榆林化学有限责任公司开展了 400 万吨/年 CCS 示范项目是我国拟建的最大的咸水层二氧化碳捕集封存项目。项目将煤化工装置排放的二氧化碳进行捕集，压缩后注入地下咸水层永久封存，并对二氧化碳地质封存、逃逸、储存能力及鄂尔多斯盆地地质特征等方面进行系统性研究。

- 预计用 10 年时间分三个阶段实施：第一阶段，率先建成 40 万吨/年 CCS 先导试验项目；第二阶段，开展百万吨级 CCS 技术开发与工程示范，建成国内第一个咸水层 CCS 地质封存实验室，牵头申报国家级 CCS 地质封存重大专项、CCS 地质封存研发中心，培养咸水层地质封存研究+应用型高端人才体系；第三阶段，建成封存规模世界前三、封存技术世界一流的咸水层 CCS 示范项目，涵盖榆林化学周边企业年排放二氧化碳约 2000 万吨，打造鄂尔多斯盆地东部区域 CCS 地质封存枢纽[41]。

陕西碳封存先导试验

- 2025 年 3 月，40 万吨/年先导试验项目榆碳 1 井相关系统顺利完成中交，并有待适时封存。

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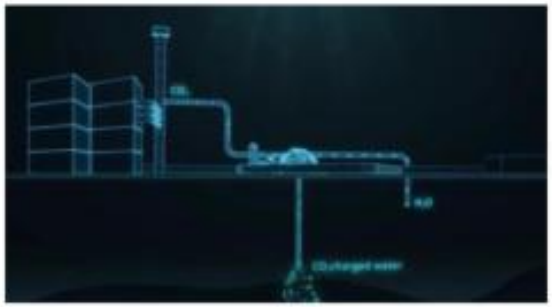
41 中国石油和化学工业联合会. 陕煤榆林化学 40 万吨/年 CCS 先导试验项目开钻[EB/OL]. (2023-04-03) [2025-03-12]. http://www.cpcif.org.cn/detail/e7cc6372-e422-480b-99bb-d90390a6f890

03.新型地质碳封存技术探索

- 玄武岩 CO_2 矿化封存是一种无需考虑盖层圈闭条件的地质碳汇模式，有望为工业碳源大规模集中处置提供更多的途径。据相关调查，我国玄武岩大体分布在东北、西北、西南等地区[42]。



玄武岩封存技术

- 我国有多个地区对玄武岩碳封存潜力进行了评价。例如，海南省开展了琼北地区玄武岩 CO_2 矿化封存潜力评价及其地质保障技术应用研究，提出了玄武岩 CO_2 矿化封存潜力评价方法，建立了琼北地区玄武岩地层 CO_2 原位矿化封存的试验性场地[43]。

- 我国也在开展玄武岩技术研究和示范工作。例如，福建省地质调查研究院与中国科学院地质与地球物理研究所的联合科研团队实施了国内的首例玄武岩原位封存二氧化碳的试验[44]；腾讯集团和冰岛碳封存公司 Carbfix 合作，正在建设国内二氧化碳玄武岩矿化封存示范项目[45]。

04.更大规模的驱油封存项目

- 2022 年 9 月，我国首个百万吨级 CCUS 驱油项目“齐鲁石化——胜利油田百万吨级 CCUS 项目”正式运行[46]。

齐鲁石化 CCUS 项目

- 中国石油推动了以松辽盆地 300 万吨 CCUS 重大示范工程为代表的“四大六小”产业格局。其中，长庆油田制定了《长庆油田千万吨级 CCUS 中长期发展规划》，总体规划部署年注气千万吨级规模的 CCUS 工程，“十四五”期间分别在陕、甘、宁三省区建设 3 个百万吨级 CCUS 工业化示范工程[47]。吉林油田建成了全产业链、全流程 CCUS 项目，预计 2025 年年注入二氧化碳能力达到 138 万吨[48]。

05.现场监测体系不断完善

- 现场监测技术体系在确保 CCUS 项目安全性、量化 CCUS 项目减排量、评价项目运行和质量等方面具有重要作用[49]。我国相关科研院所、企业正不断推动和完善 CCUS 现场监测技术体系。

- 例如，延长石油结合工程项目开展了 CO_2 动态监测技术体系相关探索[50]，通过土壤、地表及大气等方面的监测技术应用，构建了从空间上覆盖地表、土壤、浅层地下水、地下深部，从时间上贯穿 CO_2 注入前、注入过程中、注入后的一体化监测体系。

China Geological Survey (CGS)

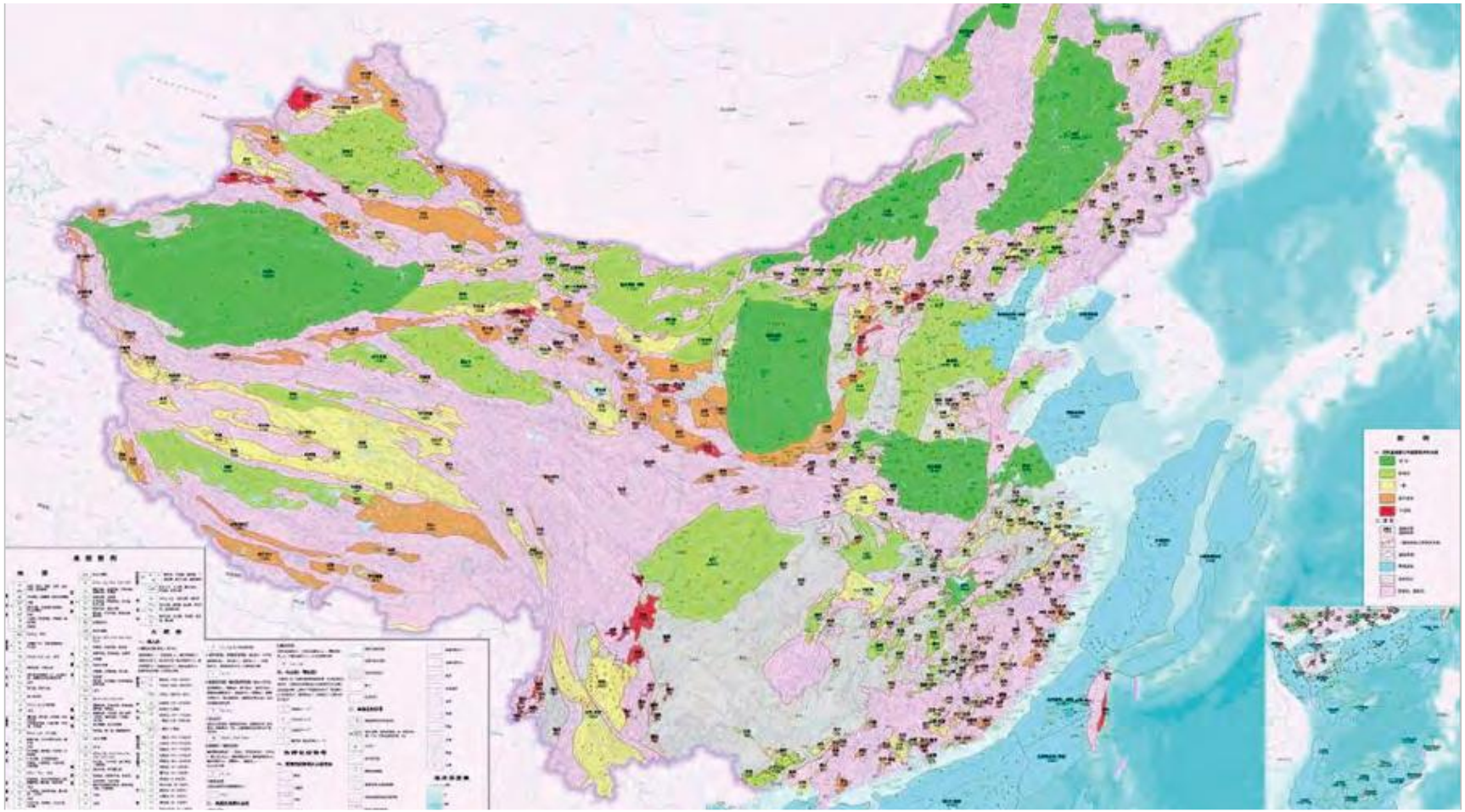
- In 2009, China Geological Survey (CGS) initiated The implement program on potential CO_2 geological storage capacity assessment in China.
- In 2009, the Ministry of Land and Resources involved the investigation and assessment on CO_2 geological storage capacity into the survey of the monitoring and evaluation of global climate change and underground space room resource assessment, which is part of geological basic information survey plan of “Geological Mineral Support Engineering Program(2010-2020)”.
- In 2010, the “Potential Capacity Assessment and Demonstration Project of CO_2 Geological Storage in China” was launched by the Centre for Hydrogeology and Environmental Geology, CGS and achieved some progress on it.

CGS objective

The objectives of the project are to

- evaluate the potential capacity for CO_2 geological storage in major sedimentary basins of China,
- delineate suitable targets for CO_2 geological storage,
- find the target sites of CO_2 geological storage based on their suitability,
- map the atlas of CO_2 geological storage suitability assessment based on sedimentary basins, and the 1:5,000,000 atlas of CO_2 geological storage demonstration projects would be implemented with industry.

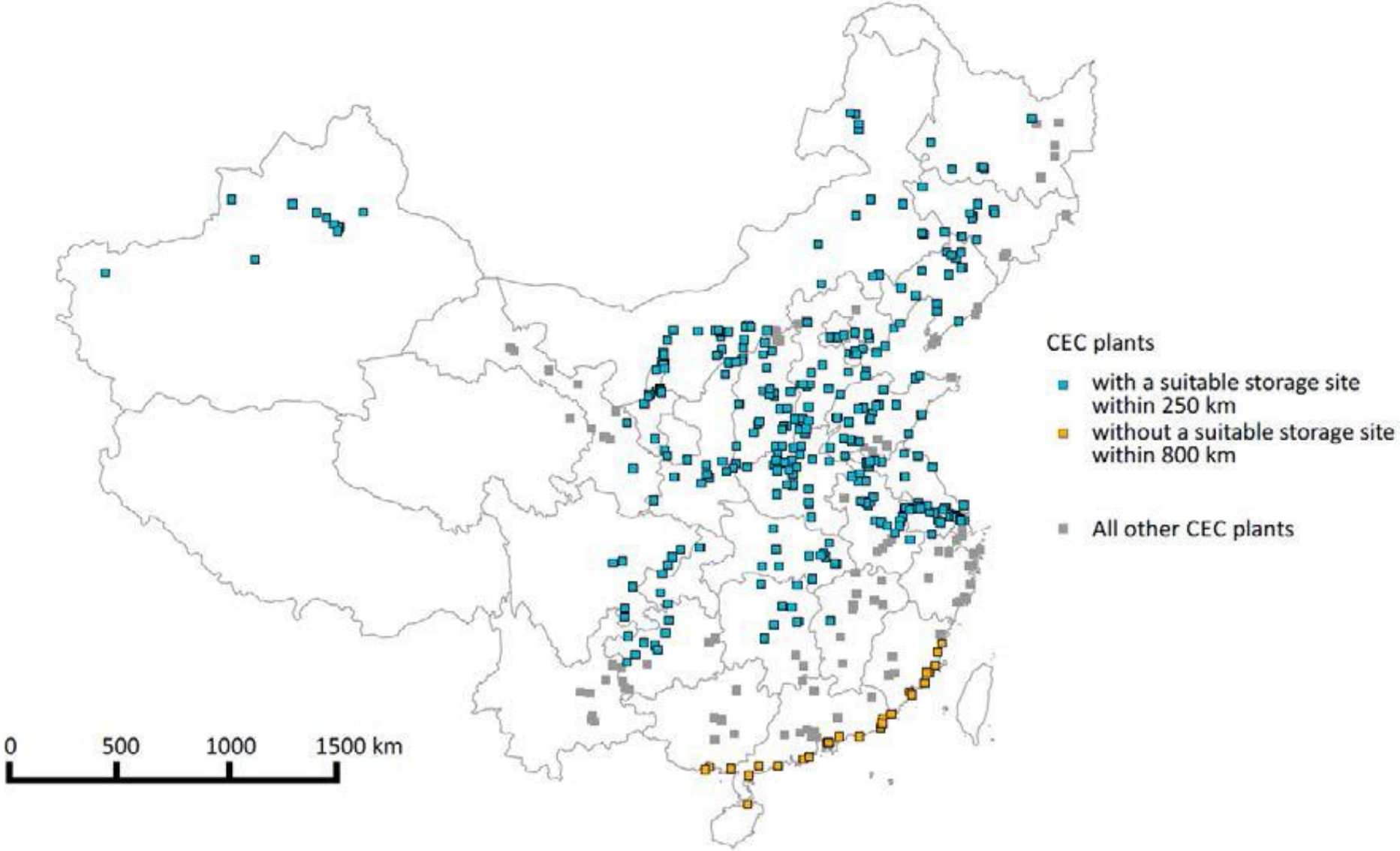
Primary assessment of CO_2 storage potential capacity in China



CGS results

- The main CO_2 geological storage site is deep saline aquifers.
- The deep saline aquifers have huge potential CO_2 storage capacity, making up 98.1% of total capacities and far exceeding that of oil field, natural gas field and coal bed methane field.
- The result reveals that the basin groups in East China own the greatest potential geological capacity, and the results of suitability assessment shows that 74 basins are moderately suitable for storage, occupying 18.97%, and 11 are suitable, occupying 2.82%.

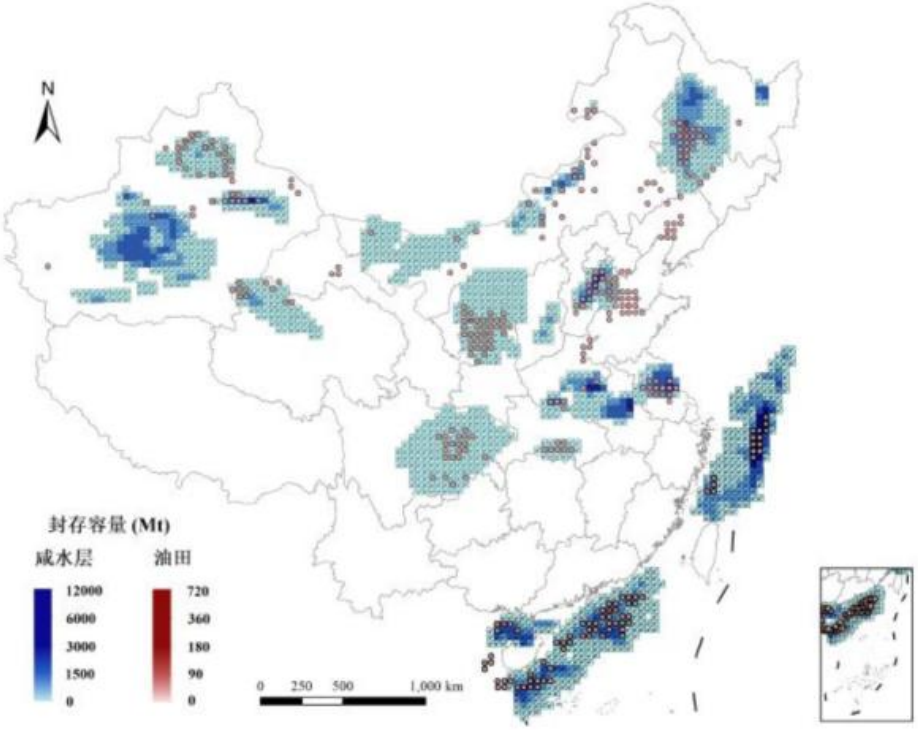
China's coal-fired power plants and their proximity to suitable CO2 storage areas



Source: IEA (2016), Ready for CCS retrofit: The potential for equipping China's existing coal fleet with carbon capture and storage

源汇匹配潜力

- 中国理论 C O 2地质封存 容量 约为1.21～4.13万亿吨，主要包括咸水层、油气田等地质构造。已探明油田可封存约200亿吨CO2，其中适宜封存的油藏容量约50亿吨CO 2。已探明气藏最终可封存约150亿吨CO2。深部咸水层的封存容量为0.16～2.42万亿吨，塔里木盆地、鄂尔多斯盆地、松辽盆地、渤海湾盆地、珠江口盆地等大中型沉积盆地，封存容量较大，条件相对较好。



中国理论地质封存容量

从源汇分布情况看，中国新疆、陕西、内蒙古等西北地区化石能源资源丰富，与塔里木盆地、鄂尔多斯盆地等陆上封存地匹配度较高。

东北、华北和川渝地区碳源与渤海湾盆地、松辽盆地、四川盆地和苏北盆地等大中型沉积盆地空间匹配相对较好。

华东大部分地区和华南地区能源消费密集， CO_2大量集中排放，陆上适合封存的盆地少、封存容量小，且受人口密集分布等影响，封存选址较为困难；可根据源汇匹配情况考虑实施海上地质封存的可行性。

Storage activity in……

- Mexico: Mexico helped develop the *North American Carbon Storage Atlas published in December 2012, and is now focused* on basin-level studies.
- Middle East and North Africa: In Salah, in Algeria, was one of the world's first operating CCS projects. Saudi Arabia actively supported the inclusion of CCS as an eligible offsetting activity under the CDM; Qatar is developing some of the world's largest CCS research initiatives.
- South Africa: The South African Centre for Carbon Capture and Storage (SACCCS) manages the country's CCS roadmap. *Atlas on Geological Storage of Carbon Dioxide in South Africa*
- Indonesia: In 2009, this 'Indonesia CCS Study Working Group' produced a comprehensive report on the potential for CCS in Indonesia.
- Malaysia: PETRONAS, the national oil company, is developing a CO2 mitigation plan that includes CCS.

CO_2 storage

Geologic carbon storage	Monitoring, verification and accounting(MVA)	Simulation and risk assessment	Regional carbon sequestration partnerships
Research and technology implementation pathways			
• Improved fundamental understanding • Technology development	• Atmospheric and remote sensing • Near-surface monitoring • Wellbore monitoring • Deep subsurface monitoring • Accounting protocols • Technology development	• Mathematical models development/verification • Improved risk assessment protocols	• Characterization phase • Validation phase • Educational and outreach
Critical challenges			
• Development of robust, flexible accounting protocols • Intelligent monitoring networks • Develop robust risk assessment process models	• Develop mitigation options for identified risks • Develop best practices • Effective public education and outreach	• Optimize well design, well management, and well integrity • Maximize CO2 injectivity and storage capacity • Long-term storage security	• Scale-up to commercial-size projects • Development of regulatory framework • Quantification/verification of stored CO2

Useful references

- DOE/NETL carbon dioxide capture and storage RD&D roadmap, 2010 December www.netl.doe.gov
- Carbon capture, utilization and storage --- technology development in China
- THE GLOBAL STATUS OF CCS
- 20 years of Carbon Capture and Storage-Accelerating Future Deployment, IEA
- 中国碳捕集利用与封存进展系列报告-新进展、新阶段、新趋势2025